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Superhydrophobic membrane: progress in preparation and its separation properties

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Abstract: Superhydrophobic membrane that is highly resistant to wetting by aqueous solution has gained great attention because of its potential to be applied in many emerging membrane processes such as membrane gas absorption (MGA) and membrane distillation (MD). Numerous approaches have been proposed to obtain membranes with superhydrophobic surface from materials with various degrees of hydrophobicity. This paper then reviews the progress in superhydrophobic membrane preparation and its separation properties. A brief description of superhydrophobicity is firstly presented. Preparation methods of the superhydrophobic membrane are subsequently reviewed, including direct processing method and surface modification of the existing membrane. Finally, the separation properties and challenges of superhydrophobic membranes are discussed. This article could provide an insight for further development of superhydrophobic membrane.

Keywords: membrane preparation; separation properties; superhydrophobic membrane; surface modification; wetting.

1 Introduction

Membrane separation performance is highly affected by its surface characteristics. There are two surface wetting

properties: hydrophilic and hydrophobic. A water droplet tends to spread on a hydrophilic membrane surface with contact angle (CA) less than 90° before being permeated through a porous membrane. On the other hand, it will minimize contact with the hydrophobic membrane surface and form a spherical shape with CA more than 90° . Generally, the former is favorable for aqueous system application while the latter is suitable for non-aqueous one. Surfaces with extreme wetting properties have been a great interest in membrane development for various separation processes (Feng and Jiang 2006, Kwon et al. 2012, 2015, Kota et al. 2014).

Membranes with high hydrophobicity are important for membrane distillation (MD) (Drioli et al. 2005, Khayet and Matsuura 2011), membrane gas absorption (MGA) (Karoo and Sirkar 1993, Klaassen et al. 2005, Mansourizadeh and Ismail 2009, Wenten et al. 2012, 2017), and other hydrophobic membrane processes (Song et al. 2004, Song and Lee 2005, Agarwal et al. 2013, Zhu et al. 2014), where the membranes are not permitted to be wetted by aqueous solution, or the permeation of non-polar compounds is desired to be promoted. For this purpose, hydrophobic polymers having low surface free energy such as polypropylene (PP), polyethylene (PE), polyvinylidene fluoride (PVDF), and polytetrafluoroethylene (PTFE) are usually used as membrane materials. However, some studies reported that these membranes were still to be wetted by some aqueous solution (Mosadegh-Sedghi et al. 2014), leading to deterioration of their performances in long-term operation. This phenomenon can be addressed by improving membrane surface hydrophobicity to produce superhydrophobic membrane. Superhydrophobic modification not only is limited to polymeric membranes but can also be applied to inorganic membranes.

A superhydrophobic surface is highly resistant to wetting. This is characterized by water contact angle (WCA) more than 150° , contact angle hysteresis (CAH), and sliding angle (SA) less than 10° (Roach et al. 2008, Celia et al. 2013). The superhydrophobic surface is inspired by lotus leaves (*Nelumbo nucifera*), which is well known for its water-repellent feature and self-cleaning effect (Lee and Michielsen 2006, Zhao et al. 2012). Investigation on the micromorphological characteristic of water-repellent plant surfaces showed that leaves with permanently water-repellent (such as *Marsilea mutica* and *N. nucifera*

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leaves) are characterized by convex to papillose epidermal cells and a very dense layer of epicuticular waxes (Neinhuis and Barthlott 1997). The surface of lotus leaf is uniformly textured with 3–10- μm sized protrusions and valleys and 70–100-nm sized particles of a hydrophobic wax-like material (Guo and Liu 2007). The combination of the binary structure and wax layer results in superhydrophobicity, i.e. a high WCA of around 162° and a low SA of about 3° . Inspired by nature, biomimetic superhydrophobic surfaces have been fabricated using various approaches (Guo et al. 2011). Artificial superhydrophobic surface fabrication was first reported in 1996 by using alkylketene dimer (Cengiz et al. 2012), and then a superhydrophobic surface was obtained from PP for the first time in 2003 (Erbil et al. 2003, Cengiz et al. 2012). Later, various alternative methods were reported, which can be divided into two main groups: (1) surface roughening of a low-surface-energy material and (2) surface roughening followed by hydrophobization (Chao-Hua et al. 2010).

The superhydrophobic membrane was developed slowly in the past, as researchers were more interested in reducing membrane hydrophobicity for permeability enhancement or fouling control in the aqueous applications (Yang et al. 2011, Ahmad et al. 2014, Himma et al. 2016). However, great attention and rapid development were given recently. As can be seen in Figure 1, the number of published papers related to “superhydrophobic membrane” increased rapidly in the recent 3 years. Since 2006, there have been many reviews on superhydrophobic surface that can be found (Feng and Jiang 2006, Ma and Hill 2006, Li et al. 2007, Kim 2008, Roach et al. 2008, Zhang et al. 2008a, Chao-Hua et al. 2010, Shirtcliffe et al. 2010, Guo et al. 2011, Ji et al. 2012, Celia et al. 2013, Motlagh et al. 2014), including the introduction of the basics and the various approaches

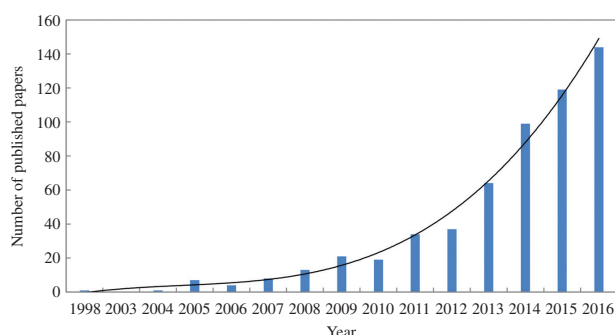


Figure 1: The number of published papers related to the search term “superhydrophobic membrane”.

Source: Web of Science [TOPIC (superhydrophobic membrane); December 2016].

to obtain superhydrophobic surface. However, review on superhydrophobic membranes, which were intended for membrane application, has not been reported elsewhere. A closely related review was reported by Ahmad et al. (2014), but it is not focused on superhydrophobicity and limited on preparation and characterization. Contributing to provide an insight for the development of superhydrophobic membrane, this article reviews superhydrophobic membrane from a different perspective. Firstly, determination of superhydrophobic terminology is introduced briefly, followed by a review of superhydrophobic preparation progress. Subsequently, the separation properties and challenges of the superhydrophobic membrane are then discussed.

2 Determination of superhydrophobicity

Membrane wetting property is affected by its material surface free energy. This is usually determined by CA, as can be seen in Young’s equation (Young 1805):

$$\gamma_{LV} \cos \theta = \gamma_{SV} - \gamma_{SL}, \quad (1)$$

where γ_{LV} is liquid/vapor surface tension, γ_{SV} is solid/vapor surface tension, γ_{SL} is solid/liquid surface tension, and θ is CA (see Figure 2A).

However, the equation is limited for smooth and homogenous surfaces (Good 1992). Wenzel (1936) then showed that surface roughness (r) has a major impact on the wetting properties, which is described by Eq. (2):

$$\cos \theta_r = r \cos \theta_e, \quad (2)$$

where r is a ratio of the wetted surface area to the projected surface area below the droplet, θ_r is CA on a rough surface, and θ_e is Young’s equilibrium CA.

The Wenzel model, which is limited to the homogeneous rough surface, was then extended by Cassie and Baxter (1944). For the porous membrane with a heterogeneous surface, air trapped in the pores makes them unfilled by the liquid.

$$\cos \theta_c = f_1 \cos \theta_e - f_2, \quad (3)$$

where θ_c is heterogeneous CA, f_1 is a fraction of the liquid area in contact with the membrane, and f_2 is a fraction of the liquid area in contact with air in the porous membrane. If $f_1 + f_2 = 1$, hence,

$$\cos \theta_c = f_1 \cos \theta_e + f_1 - 1. \quad (4)$$

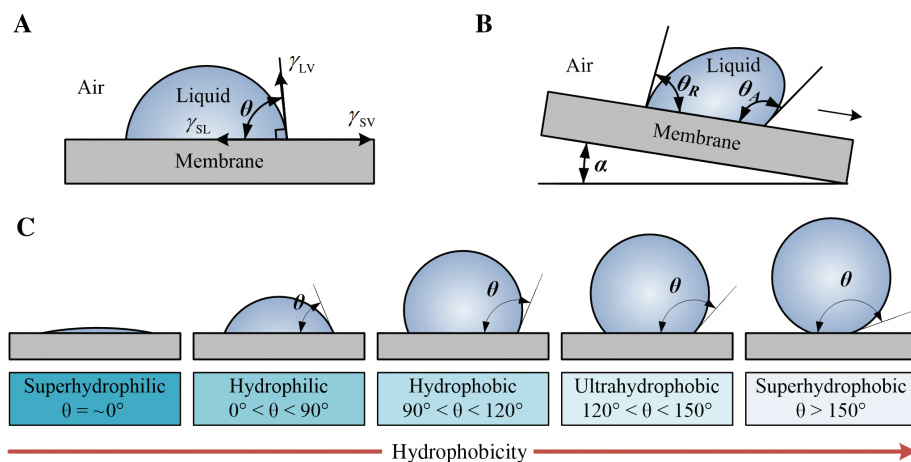


Figure 2: Parameters characterizing the hydrophobicity of the membrane surface.

(A) Static contact angle which is the tangent angle of the liquid-vapor interface at the three phase contact line. It is influenced by surface tensions acting at the three-phase contact line. (B) Contact angle hysteresis and sliding angle which are obtained by tilting the membrane surface. (C) The contact angle of liquid droplets on the membrane surface with various wetting properties. The higher the contact angle, the more hydrophobic the membrane surface is.

According to Eq. (4), CA θ_c will increase with the decrease of f_1 . This implies that enhancement of membrane surface roughness, which increases the air fraction and decreases the contact area between the liquid and the membrane, will improve the hydrophobicity of the surface.

In a further study, a general form of stability criterion was defined to determine if the Wenzel or Cassie-Baxter wetting mode is favorable on a given surface, given by Milne and Amirfazli (2012):

$$\cos \theta_{cr} = \frac{-f_2}{r - f_1}, \quad (5)$$

where θ_{cr} is the critical intrinsic CA. If $\theta_e > \theta_{cr}$, Cassie-Baxter mode is favorable, whereas if $\theta_e < \theta_{cr}$, Wenzel mode is instead favorable. Wenzel and Cassie-Baxter models have been used extensively in studies of the superhydrophobic surface (Roach et al. 2008, Bittoun and Marmur 2009) despite the fact that some researchers argue on validity (Erbil 2014). It is suggested that there are some criteria to be considered to obtain a correct prediction.

To determine hydrophilicity/hydrophobicity, WCA of 90° is usually the red line (Quééré 2008). When the WCA is between 0° and 90° , the surface is called hydrophilic. Usually, a surface is described as ultrahydrophobic when the WCA is above 120° (Roach et al. 2008) and superhydrophobic when the WCA is more than 150° (Roach et al. 2008, Drelich et al. 2011). CA of liquid (water) droplets on a membrane surface with various wetting properties is shown in Figure 2C.

In the case of the superhydrophobic surface, besides WCA, superhydrophobicity is also indicated by relatively low CAH and SA (Roach et al. 2008, Celia et al. 2013). Figure 2B shows CAH and SA on the surface of a tilted membrane. CAH is the difference of advancing CA (θ_A) and receding CA (θ_R). The low CAH means that the water droplet can roll off easily. SA (α) is defined as a critical angle where liquid droplet with a certain weight starts to slide down on the tilted plane. Usually, SA value drastically decreases as WCA increases, which can be correlated with an increase in surface roughness (Miwa et al. 2000). Hence, a surface is called superhydrophobic when its WCA is more than 150° with low CAH and SA of less than 10° .

In addition to surface roughness and hydrophobicity or hydrophilicity, the CA value is influenced by pore size and porosity (Hebbar et al. 2017). As reported by Li et al. (2014a,b,c), the CA of liquid in a glass pore increases with the decrease of pore size. Thus, although the membrane is more hydrophobic, the CA value may be lower when the membrane pore size is bigger. In addition to superhydrophobicity, the effect of pore size on membrane wettability should also be taken into account in the development of superhydrophobic membrane.

3 Preparation of superhydrophobic membrane

It is well recognized that hydrophobicity can be improved by controlling surface chemical and morphology or

surface roughness. The minimum surface free energy that can be obtained on a smooth surface terminated by $-\text{CF}_3-$ is 6.7 mJ/m^2 corresponding to WCA of 119° (Nishino et al. 1999). Therefore, to obtain superhydrophobic surface, it is necessary to combine both low surface free energy material and surface roughness. Solution casting, sol-gel processing, etching, chemical vapor deposition (CVD), chemical and electrochemical depositions, and electrospinning are ways to make rough surfaces (Ma and Hill 2006). For hydrophobization, reactive molecules such as alkyl chain fatty acids, polydimethylsiloxane, or other hydrophobic polymers are commonly used (Chao-Hua et al. 2010). A different number of steps to prepare superhydrophobic membrane may be required, including direct processing of the membrane or membrane surface modification using one-step or two-step method (Crick and Parkin 2010, Oliveira et al. 2012). In the one-step method, the superhydrophobic surface can be prepared by surface roughening of or with low-surface-energy material. In the two-step method, the superhydrophobic polymeric membrane is prepared by surface roughening followed by hydrophobization using low-surface-energy material or hydrophobization using low-surface-energy material followed by roughening the surface.

Superydrophobization can be introduced for organic (polymeric) and inorganic membranes. For polymeric membrane, a direct roughening process can be conducted if the polymer used is hydrophobic. But both surface roughening and hydrophobization using low-surface-energy material are required if the polymeric membrane is produced by using hydrophilic material. The modification can be achieved during membrane preparation process or by surface modification of the existing membrane. Meanwhile, for the inorganic membrane, surface modification is the most feasible way because of its hydrophilic nature.

3.1 Direct processing method

In order to enhance membrane hydrophobicity during the process of membrane preparation, blending is a simple method that has been quite widely used. For this purpose, surface modifying macromolecule (SMM) is often used as an additive in the polymer solution (Figure 3B). Hydrophobic SMM is macromolecule with an amphipathic structure, consisting of the hydrophilic and hydrophobic component. During membrane formation, the hydrophobic SMMs migrate to the surface due to the tendency to minimize the interfacial energy (Hamza et al. 1997). SMM migration depends on SMM properties and the

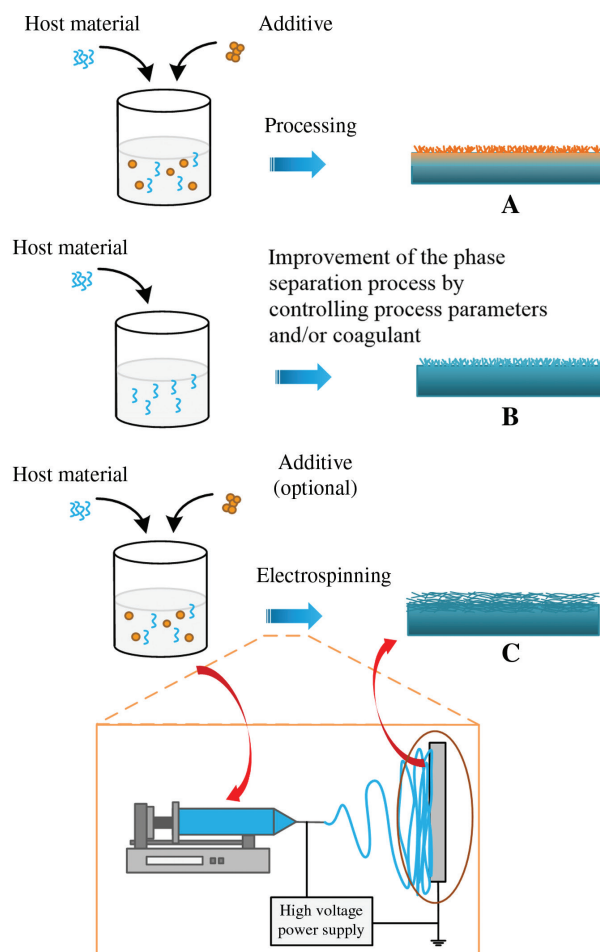


Figure 3: Hydrophobic and superhydrophobic membrane directly prepared through blending with hydrophobic material (A), improvement of phase separation process (B), and electrospinning process (C).

conditions of membrane preparation (Bakeri et al. 2012a, Mansourizadeh et al. 2014). One of the most effective parameters is evaporation time (i.e. time between casting and immersion in the coagulation for case of flat sheet membrane, or residence time in air gap for case of hollow fiber membrane), as it means the time allowed for the SMM migration to the surface (Bakeri et al. 2012a,b, 2013, Mansourizadeh et al. 2014).

A fluorinated SMM was blended into casting solution of polyetherimide (PEI) to fabricate a composite porous flat-sheet membrane through phase inversion technique (Essalhi and Khayet 2012). The WCA of the top surface was 94.4° , which was higher than that of the bottom surface (78.6°), indicating the SMM migration to the top surface. The use of such method by using another host polymer of polyethersulfone (PES) has also been reported (Suk et al. 2010). SMM has also been used in the fabrication of

surface modified hollow fibers, which the host polymer included PEI (Bakeri et al. 2012a,c), polysulfone (PSf) (Rahbari-Sisakht et al. 2012a), and PVDF (Rahbari-Sisakht et al. 2012b, Mansourizadeh et al. 2014). SMM blending increased the WCA of the modified PSf, PEI, and PVDF membranes to 62–73° (Rahbari-Sisakht et al. 2012a), 85–89° (Bakeri et al. 2012a,c), and 83–96° (Rahbari-Sisakht et al. 2012b, Mansourizadeh et al. 2014), respectively.

It can be pointed out from the available literature that SMM blending is inadequate to render superhydrophobicity, which can be ascribed to insignificant enhancement on membrane surface roughness. Nanoparticles blending, especially superhydrophobic nanoparticles, in hydrophobic polymer solution should highly increase the surface roughness, leading to superhydrophobicity. Rezaei et al. used nano-sized hydrophobic montmorillonite (MMT) particles as fillers in PVDF and PEI matrix, which increased the membrane porosity, which was not exhibited in SMM blending (Rezaei et al. 2014a,b, 2015, Rezaei-DashtArzhandi et al. 2015). However, the WCAs were below 100°. Moreover, a study by Efome et al. (2015) showed that blending superhydrophobic SiO₂ nanoparticles into the PVDF dope solution did not result in superhydrophobic membrane. This might be attributed to an inappropriate surface roughness due to the small migration of nanoparticles to the membrane surface.

Other direct processing methods that have succeeded in producing superhydrophobic membranes are the improvement of phase separation process and electrospinning method, which are then reviewed below.

3.1.1 Improvement of phase separation process

Membrane hydrophobicity can be enhanced by increasing the surface roughness through improvement of the phase separation process. This can be done by controlling process parameters, polymer solution, and coagulant parameters (Figure 3A). Although phase-inversion techniques, such as non-solvent-induced phase separation (NIPS), evaporation-induced phase separation (EIPS), vapor-induced phase separation (VIPS), and thermally induced phase separation (TIPS), could be used for this method (Ulbricht 2006), there are only few works reporting superhydrophobic membrane preparation using this method.

Kuo et al. (2008) prepared PVDF microporous membrane by the NIPS process. Using alcohol as the coagulant (non-solvent), delay in demixing phase separation occurred and a porous structure was formed resulting in a highly hydrophobic microporous membrane (WCA ~ 148°). Peng et al. (2012a) have reported that phase

separation in the VIPS process could be controlled by changing the exposure time. It has been revealed that long exposure time gave sufficient time for the polymer to crystallize. This resulted in the formation of porous skin and small crystalline particle morphology, leading to an increased membrane surface hydrophobicity. Besides, initial polymer content in the casting solution and the air temperature could affect the surface morphology and the hydrophobicity of membranes (Peng et al. 2012b). It was shown that low PVDF contents (2–4 wt%) and high air temperature (above 60°C) led to a net-like surface morphology through spinodal decomposition, resulting in a superhydrophobic surface.

3.1.2 Electrospinning

Electrospinning enables us to produce micro- and nanoscale fibers using either electrostatically charged polymeric solutions or melt (Wang et al. 2011). A superhydrophobic nanofibrous membrane can be directly produced due to the hydrophobicity of polymer used and the surface roughness introduced during spinning (Figure 3C). To obtain electrospun fibers with desired morphology and properties, both system parameters (such as polymer, solvent, viscosity, surface tension, conductivity, and dielectric properties of the solution) and process parameters (such as mass flow rate, electrostatic field, distance between tip of capillary tube to collector screen, temperature, humidity, and air velocity in the electrospinning chamber) need to be controlled (Khan et al. 2013).

Kang et al. (2008) fabricated electrospun membrane using polystyrene (PS) solution with different solvents, i.e. chloroform, N,N-dimethylformamide (DMF), and tetrahydrofuran (THF). The membrane obtained using a non-volatile solvent (DMF) had a protuberant secondary structure on its surface, leading to superhydrophobicity with a WCA of $154.2 \pm 0.7^\circ$.

The viscosity of polymer solution affects the fibers dimension and membrane pore size. As reported by Liao et al. (2013a) in fabrication of PVDF nanofiber membranes by electrospinning, diameters of nanofibers spun by a 8-wt% PVDF solution (viscosity of 1.00 Pa·s) were in the range 1–1.5 μm while the diameters of nanofibers spun by a 7-wt% PVDF solution (viscosity of 0.50 Pa·s) were in the range of 0.2–0.5 μm. The former membrane had a maximum pore size and a mean pore size of 1.72 μm and 0.91 μm, respectively, whereas the latter membrane had a maximum pore size and a mean pore size of 1.12 μm and 0.75 μm, respectively. These results suggest that thinner nanofibers of the membrane with smaller pore

size can be obtained by low viscosity (low concentration) of the polymer solution. Patel and Chase (2014) used PP concentrations of 2, 3, and 4 wt%, and it resulted in fibrous membrane with fiber diameters of 876, 1082, and 1710 nm, respectively. The fibrous membrane with smallest fiber size (876 nm) had the best superhydrophobicity (the highest WCA of 165° and the lowest hysteresis of 6°).

The important process parameters such as electrical field, solution flow rate, and the distance between tip to collector also influence the electrospun fiber diameter and the formation of beads and pores, which would be of interest in the field of superhydrophobic surfaces (Sas et al. 2012). So far, published works investigating the effect of those parameters on superhydrophobicity are still very rare. A study by Sarkar et al. (2011) reported that different flow rates in electrospinning process resulted in different WCA and SA, which was attributed to the particle size, bead number, and fiber uniformity. Using 5% PVDF solution and varying the flow rate of 0.43 ml/h, 0.26 ml/h, and 0.15 ml/h, the highest WCA of about 144° and the lowest SA of about 24° were achieved when using a flow rate of 0.26 ml/h. Liu et al. (2016) have reported that different voltages resulted in different WCAs. However, the success to get superhydrophobic membrane simply by controlling the individual process parameters has not been reported. As shown by Ren et al. (2017), the slight increase of WCA from 156.5° to 160.5° was obtained when the voltage increased from 10 to 11 kV, which the high WCA above 150° was primarily due to the optimum electrospinning solution used.

Electrospinning can also be used for inorganic-polymer composite nanofibers preparation. The use of nanoparticles can increase the surface roughness that is favorable to hydrophobicity of the membrane surface. Wang et al. (2011) used epoxy-siloxane modified SiO_2 nanoparticles, which were introduced into PVDF solution. As the mass ratio of modified SiO_2 nanoparticles/PVDF increased to 5.00:1, WCA improved to 161.2° and water SA decreased to 2.17° . Liao et al. (2014b) prepared composite nanofiber membranes using two modified silica-blended PVDF dopes. Silica particles added in the first dope solution were smaller with diameters of 5–10 nm, while in second dope, larger hydrophobic SiO_2 particles with aggregate diameters of 0.2–0.3 μm were used. Both composite nanofiber membranes had high WCA above 150° and excellent water repellency. Other than SiO_2 nanoparticles, ZnO nanoparticles have been used in superhydrophobic composite nanofiber membranes (Liu et al. 2016). Li et al. (2015) have fabricated electrospun composite fibrous membranes using a mixture of polyurethane (PU), terminal fluorinated polyurethane (FPU), and carbon nanotubes (CNTs). In addition to superhydrophobicity,

the fibrous membrane exhibited high waterproofness and good breathability. Recently, Tijing et al. (2016) used CNTs, which were incorporated into polyvinylidene fluoride-co-hexafluoropropylene (PcH) solution. The presence of CNTs produced beads on the surface of nanofiber membrane, which increased the surface roughness. Due to the increased surface roughness and hydrophobicity of CNTs, WCA increased to 158.5° with the incorporation of 5-wt% CNTs in the nanofiber. Besides, CNT incorporation improved the mechanical properties of the composite membrane, which is needed for long-term operation in a separation process. The properties of superhydrophobic membranes resulting from electrospinning process with varied conditions are summarized in Table 1.

3.2 Surface modification

3.2.1 Plasma treatment

Plasma treatment is considered to be a feasible method for membrane surface modification without affecting the bulk properties. Plasma treatment can modify a membrane surface through surface reactions, plasma polymerization, or physical etching, depending on the gas type used (Ebnesajjad 2014). Regarding hydrophobic enhancement, fluorine-containing plasma gases are mostly used. Bae et al. (2001) have characterized the surface of microporous PP membranes modified by plasma treatment. It has been shown that from four different reagents (acrylic acid, allylamine, water, and Freon-116), Freon-116 was the only reagent improving the membrane hydrophobicity due to the presence of C-F bonding. Besides Freon-116, other reagents that have been used include CF_4 (Lin et al. 2009, Wei et al. 2012, Yang et al. 2014, 2015, Chen et al. 2017); PTFE (Youngblood and McCarthy 1999, Franco et al. 2011a,b); 1,1,1,3,3,3-hexafluoro-2-propanol (HFIP); and hexamethyldisiloxane (HMDSO) (Tur et al. 2012).

Lin et al. (2009) have used CF_4 to modify PP hollow fiber membrane through plasma treatment. Membrane hydrophobicity was improved with increasing WCA from 115° to 143° after plasma treatment for 5 min (power at 20 W). This is due to the increase of fluorine-to-carbon (F/C) ratio and surface roughness. CF_4 plasma has also been used to modify asymmetric hydrophilic PES membranes (Wei et al. 2012). After plasma treatment, the WCA at the top surface increased from 58° to 124° and the WCA at the bottom surface increased from 0° to 125° . Fluorination was ascribed as the main cause for the conversion of hydrophilic membranes into ultrahydrophobic ones. Superhydrophobic PVDF flat sheet membranes have been successfully prepared via CF_4

Table 1: Properties of superhydrophobic membranes produced by electrospinning.

Polymer solution	Electrospinning conditions	Membrane thickness (μm)	Fiber diameter (nm)	Mean pore size (μm)/porosity (%)	WCA ($^\circ$)	Reference
PS (35 wt%) in DMF	FR = 150 $\mu\text{L}/\text{min}$ V = 16 kV TCD = 21 cm	–	–	–	PS/DMF: 154.2	Kang et al. (2008)
sPP (2, 3 and 4 wt%) in cyclohexanone/acetone/DMF (8:1:1)	FR = 25 ml/h V = 25 kV TCD = 25 cm	–	PP (2) = 876 PP (3) = 1082 PP (4) = 1710	PP (2) = –/0.94 PP (3) = –/0.94 PP (4) = –/0.95	PP (2) = 165 PP (3) = 158 PP (4) = 157	Patel and Chase (2014)
PVDF/PFOTES or SFPVDF/PFOTES (18 wt%) in DMAc	FR = 1 ml/h V = 15 kV TCD = 15 cm	–	–	–	PVDF/PFOTES = 156 SFPVDF/PFOTES = 152	Chen and Kim (2009)
PVDF (20 wt%) in DMF; with modified SiO_2/PVDF	TCD = 15 cm V = 15 kV TCD = 14 cm	–	–	–	SiO_2/PVDF (1.25:1) = 150.4 SiO_2/PVDF (1.67:1) = 156.1 SiO_2/PVDF (2.50:1) = 160.7 SiO_2/PVDF (5.00:1) = 161.2	Wang et al. (2011)
PS (20 wt %) in DMF	FR = 180 $\mu\text{L}/\text{h}$ V = 5.0 kV	–	317	–	155	Lee et al. (2013)
PVDF-HFP in acetone (10 wt%)	FR = 15 ml/h V = 30 kV	–	334	–/73	170.3	Patel et al. (2013)
PVDF (15 wt%) in DMAc/acetone (5:5); with 12 wt% PTFE	TCD = 15 cm FR = 0.5 ml/h V = 18 kV TCD = 15 cm H = 50%	98	580	0.49/69	152.2	Dong et al. (2014)
PVDF (3–5 wt%) in DMF; with silica/PVDF at 2:1; with nanofibrous (N) or nonwoven (W) support	FR = 0.02 ml/min V = 28 kV TCD = 12 cm Loop number = 6	#3S-N = 72 #5S-W = 76	–	#3S-N = 0.32/– #5S-W = 0.36/–	#3S-N = 154 #5S-W = 150	Liao et al. (2014a)
PCH (15 wt%) in DMF/acetone (4:1); with 5 wt% CNTs	FR = 0.5 ml/h V = 20–22 kV TCD = 20 cm	81	257	0.29/84	158.5	Tijing et al. (2016)

CHCl₃, Chloroform; DMAc, N,N-dimethylacetamide; FR, flow rate; H, humidity; PCH, fluoride-co-hexafluoropropylene; SFPVDF, grafted PVDF (i.e. PVDF-g-PTMSPMA); SFPVDF/PFOTES, fluorinated silane functionalized PVDF; PVDF-HFP, poly(vinylidene fluoride-co-hexafluoropropylene); sPP, syndiotactic polypropylene; TCD, tip-to-collector distance; V, voltage.

plasma surface modification with high WCA of 162.4° after plasma treatment time of 15 min and power of 150 W (Yang et al. 2014) and 162.5° after plasma treatment time of 20 min and power of 150 W (Yang et al. 2015).

Youngblood and McCarthy (1999) used simultaneous etching of PP and sputtering of PTFE to obtain a highly hydrophobic surface. The sputtering is a deposition method, in which the target material (PTFE film) is bombarded by energetic particles of sputtering gas (argon) in a radio frequency plasma reactor and then their atoms are ejected to the substrate (PP film). It was found that the roughness of PP was highly affected by plasma etching time. Later, Franco et al. (2011a,b) studied the use of PP membranes that had been sputtered with PTFE for carbon dioxide absorption. The argon flow rate, treatment time, and radio frequency power were varied to determine optimum conditions for PTFE sputtering. The WCA of PTFE-sputtered PP membrane was 26° and 12° higher than untreated PP and PTFE, respectively. Moreover, the modified membrane property could still be well maintained after MEA exposure (Franco et al. 2011a).

According to a previous investigation, time of plasma treatment and working power are the most important process parameters affecting surface hydrophobicity. As shown in Figure 4, WCAs generally increase with plasma treatment time but remain nearly constant after some minutes of plasma treatment (Bae et al. 2001, Lin et al. 2009, Yang et al. 2014) and even fall slightly for longer treatment time (Franco et al. 2011a). The increase of CA indicated that the membrane surface is more hydrophobic via formation of C-F bonds. However, further plasma treatment might hardly affect the hydrophobicity of the membrane even though more fluorine accumulated on the membrane surface, which can be attributed to the formation and breakage rates of C-F bonds being equal (Lin et al. 2009) or the formation of a certain level of C-F bonds

(Bae et al. 2001). It was reported that WCA increased with the power until it peaks at a certain power; above it, WCA dropped dramatically, in which the membrane surface becomes smoother and less porous because the deposited film has increased in thickness (Franco et al. 2011a). In addition, excessive plasma treatment could make the membrane surface charred (Lin et al. 2009). Hence, optimization of process conditions is important to obtain superhydrophobic membrane using plasma treatment.

During plasma treatment, deposition and/or etching may occur. The processes highly affect the surface roughness, which is very important in tuning surface wettability (Youngblood and McCarthy 1999, Li et al. 2016a,b), as discussed above. In addition, the processes may also alter the membrane pore size and porosity, which in turn affect the membrane permeability. As shown in Figure 4, the membrane pore size slightly increased with plasma treatment (Yang et al. 2014). Meanwhile, other studies reported that there was no pores deterioration observed even at 40 min of treatment (Bae et al. 2001), and there was a small decrease in membrane porosity (Franco et al. 2011a). The increase of membrane size and porosity can be due to the etching process, whereas the decrease can be attributed to the deposition effect. However, the important point is that the change of membrane pore structure is not significant, thus a high separation performance, thanks to superhydrophobicity without offsetting by deterioration in membrane pore structure, is expected.

Fluorine-plasma treatment is often used to improve hydrophobicity of membrane surface as presented above. Lee et al. (2011a) have reported plasma treatment to render superhydrophobicity using non-fluorinated hydrocarbon gases at ambient conditions. C6 hydrocarbons with different structure were tested, including n-hexane (linear aliphatic), cyclohexane (cyclic aliphatic), and benzene (aromatic), and it is found that the film morphology and hydrophobicity highly depend on the structure of the reagent hydrocarbon. The resulting films had smooth surface morphology and WCA of $90\text{--}95^\circ$ when n-hexane and cyclohexane were used, whereas the benzene-deposited films consisted of fine nanoparticles and exhibited superhydrophobicity with a high WCA of 167° .

Plasma treatment can also be used as pretreatment for surface conditioning before subsequent coating with low energy material such as fluorosilane. As shown in some works, argon plasma followed by modification with 1H,1H,2H,2H-perfluorodecyltrimethoxysilane has successfully imparted superhydrophobicity on the PS surface (Song et al. 2010, 2013). Rather than individual use of an argon plasma that solely gave the effect of sputtering mechanism, a combined oxygen/argon plasma was also

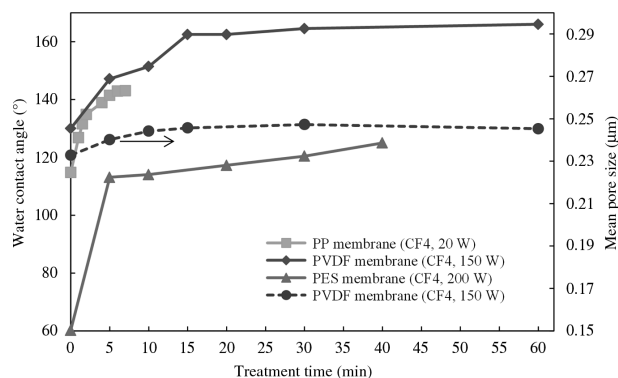


Figure 4: WCA of modified membrane as function of plasma treatment time (Lin et al. 2009, Wei et al. 2012, Yang et al. 2014).

used by West et al. (2016) as the initial treatment, which also changed the surface chemistry by imparting polar groups, so that the subsequent fluoroalkylsilane coating could be conformed and well-adhered. However, the application of such method for superhydrophobic modification of membrane surface has not been reported.

3.2.2 Deposition of rough polymer surface

In this method, a superhydrophobic membrane is obtained by depositing a rough polymeric layer on the existing membrane surface. Here, the roughness of the polymeric coating layer is created by the addition of non-solvent or incorporation of nanoparticles into the polymer solution, which is reviewed in the following sections.

3.2.2.1 Solvent/non-solvent method

The rough polymeric coating layer is prepared by the following steps. A hydrophobic polymer was firstly dissolved in a suitable solvent to form a homogeneous solution. One type or more of non-solvent was then added into the polymer solution. The resulting solution was then cast onto a membrane surface. Finally, the solvent was evaporated. The basic concept is the non-solvent's ability to aggregate the nano- and micro-scale papillae, leading to improvement of surface roughness and thus hydrophobicity (Ahmad et al. 2013c). This method basically is a phase separation method, but the improved phase separation is intended for the polymeric coating layer, not the membrane (substrate) such as in the direct processing method.

Superhydrophobic coating using solvent/non-solvent method was described at the first time by Erbil et al. (2003) using isotactic polypropylene (i-PP), but it was not applied on membrane substrate. The surface roughness was controlled by a suitable selection of solvents and temperature. A coating with the largest WCA of 160° and the best homogeneity was obtained from a solution (20 mg/ml) using methyl ethyl ketone (MEK) non-solvent at drying temperature of 70°C . The coating had smaller i-PP crystallites and narrower cylindrical bridges with enlarged pores and pore size distribution. In the same polymer/solvent system of i-PP/xylene, the use of acetone as non-solvent also could lead to the formation of bird-nest-like micro/nano-binary structures on the surface, resulting in superhydrophobicity (Ji et al. 2011). Such structures make the air fraction in the pores that are in touch with the water drop larger, resulting in an increase of WCA.

In a similar way to Erbil's work, Lu et al. (2004) fabricated superhydrophobic surface from low-density polyethylene (LDPE), in which the crystallization behavior of

LDPE was controlled by adjusting the crystallization time and nucleation rate. Cyclohexanone was used as non-solvent. The resultant surface had a floral-like crystal structure comprising hierarchical micro- and nanostructures, leading to a high WCA of $173.0 \pm 2.5^\circ$. Then the solvent/non-solvent method was used to prepare superhydrophobic surface using amorphous polymer, poly(vinyl chloride) (PVC) (Li et al. 2006). The PVC surfaces obtained using 2:1 (v/v) ethanol/ H_2O , 5:1 (v/v) acetone/ H_2O , and 5:1 (v/v) acetone/ethanol had WCA of $155.3 \pm 1.9^\circ$, $163.5 \pm 2.1^\circ$, and $163.6 \pm 1.8^\circ$, respectively. Superhydrophobic PS was also successfully obtained using ethanol as non-solvent (Yuan et al. 2007). Many uniform micro-spheres of 1–4 μm were observed on the surface structure, resulting in high WCA and low SA of $156 \pm 1.9^\circ$ and $2 \pm 1.9^\circ$, respectively.

Franco et al. (2008) then applied this method to modify porous PP flat membrane. Modification using MEK resulted in higher WCA than that using cyclohexanone. A high WCA of 169° was obtained by using MEK as non-solvent and PP concentration of 179 mg/ml, which was 42° higher than that of an untreated PP membrane and 30° higher than that of a PTFE membrane. Using the method, Lv et al. (2012a) modified PP hollow fiber membrane. The mixture of cyclohexanone and MEK system (mass ratio of 1:1) was considered as the best non-solvent in terms of hydrophobicity, coating homogeneity, and modification process efficiency. The modification improved the stability and maintained the superhydrophobicity of PP hollow fiber membrane contacting with the absorbent.

Besides, by drop-coating (Erbil et al. 2003) and spin-coating (Franco et al. 2008, Lv et al. 2012a,b), the roughened polymer coating can be deposited onto membrane surface by immersion or dip-coating. This technique is more convenient to be applied for a curved surface such as hollow fiber membrane. Moreover, it is easy to scale up and offers a good control on thickness (Grosso 2011). Ahmad et al. (2013a) have reported the preparation of superhydrophobic coating on PP hollow fibers using simple dip-coating. PP hollow fiber was immersed in non-solvent solution followed by LDPE solution. A superhydrophobic surface with the highest WCA of 161° was achieved on the PP at 20 mg/ml polymer (LDPE) concentration with a mixture of 1:1 (v/v) acetone/ethanol as non-solvent. A porous superhydrophobic layer of LDPE was also successfully dip-coated onto the PVDF hollow fibers (Ahmad et al. 2013b). Membrane surface with high WCA of $152 \pm 3.2^\circ$ was obtained using a mixture of 5:1 (v/v) acetone/ethanol as the non-solvent. The use of solvent/non-solvent approach with immersion technique was also reported by Wang et al. (2017). Propylene glycol was used as the non-solvent in PVDF solution deposited onto PVDF

hollow fiber membranes. By controlling the coating solution and immersion time, superhydrophobic surface with a high WCA of 156.8° was obtained.

In our previous work (Himma et al. 2017b), the effects of the solution and process parameters in preparation of superhydrophobic PP hollow fiber membrane using two-step dip-coating process were investigated. The results showed that the number of dipping cycle and withdrawal velocity slightly affected WCA value. Modified membrane surfaces with high hydrophobicity could be obtained by controlling the polymer solution concentration and drying temperature. Meanwhile, the coating surface homogeneity highly depended on the concentration and temperature of the polymer solution. The non-solvent used also affected the surface morphology of the modified membrane, thus affecting hydrophobicity (Himma et al. 2017a). As shown in Figure 5, it is proven that the modified membrane with an enhanced surface roughness exhibited a higher WCA.

In the coating method, controlling the coating thickness is crucial. This is because the increase in membrane thickness leads to higher mass transfer resistance (Lv et al. 2012a). However, decreasing film thickness lowers

the probability to get superhydrophobic surface (Rioboo et al. 2008). Hence, an optimum thickness should be maintained.

3.2.2.2 Incorporation of inorganic nanoparticles

Incorporation of nanoparticles in the polymer solution can induce the formation of surface with micro-nano binary roughness. Thereof, coating with the nanocomposite is widely used in superhydrophobic surface preparation (Guo and Wang 2010, Momen and Farzaneh 2012, Yu and Li 2016). In this method, good dispersion of nanoparticles in the polymer matrix is crucial, as the exposure of nanoparticles on the surface of nanocomposite coating could deteriorate the superhydrophobicity if the nanoparticles used is hydrophilic (Liu et al. 2010, Hejazi et al. 2014). Thus, nanoparticles loading is needed to be optimized. Otherwise, hydrophobic nanoparticles are used, or surface modification of the particles using hydrophobic materials such as stearic acid (Qing et al. 2015) and fluoro-silane (Hancer and Arkaz 2015) can be an alternative.

Zhou et al. (2014) used casting method to produce SiO_2 /polydimethylsiloxane superhydrophobic composite

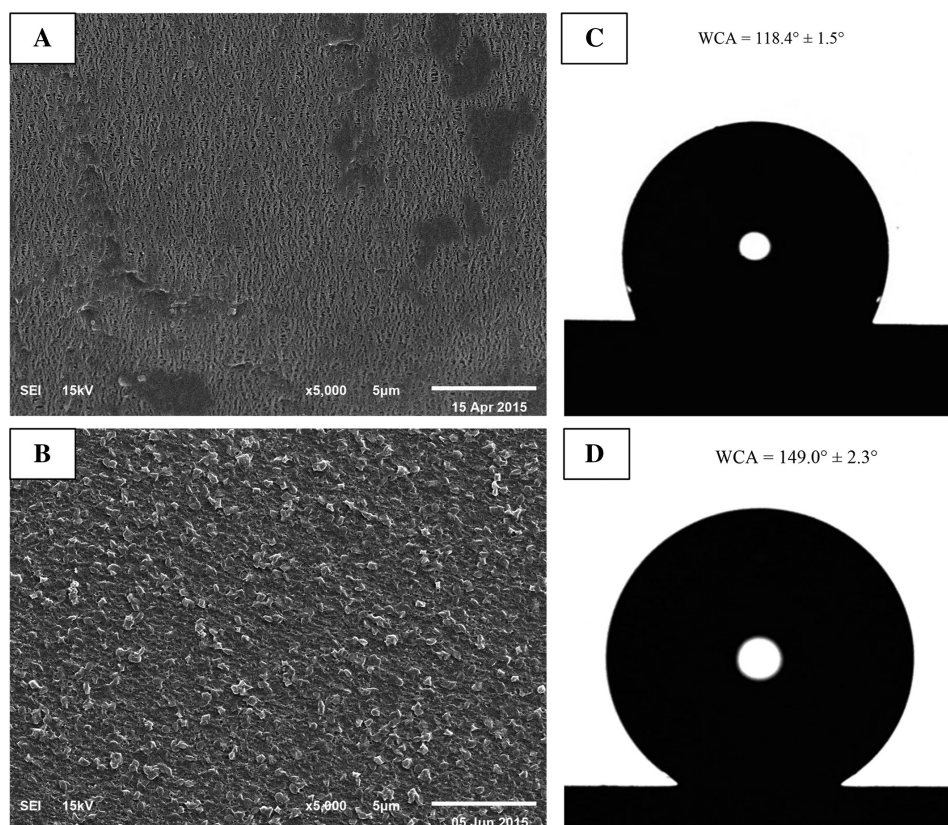


Figure 5: Microscopic images of unmodified membrane (A) and modified membrane with PP solution and MEK (B), followed by a water droplet profile on their surfaces (C and D).

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coating for pervaporation membrane with 152° and 5° in WCA and SA, respectively. The separation factor obtained using the superhydrophobic membrane was nearly five times higher than that obtained using a non-superhydrophobic membrane. Zhang et al. (2013b) prepared superhydrophobic membrane by spraying the mixture of polydimethylsiloxane (PDMS) and hydrophobic SiO_2 on PVDF membrane. The WCA and the liquid entry pressure (LEP_w) of the membrane were 156 and 275 kPa, respectively. Nanocomposite coating deposition can also be conducted by simple dip-coating (Ju et al. 2015). In Ju et al.'s (2015) work, dip-coating with nanocomposite containing hexamethyldisilazane-modified SiO_2 on the PVDF membrane increased WCA up to 158° .

Besides, the combined use of a proper solvent/non-solvent system and nanoparticles has been proposed to substantially reduce the non-solvent and nanoparticles required to achieve superhydrophobicity (Hejazi et al. 2015, Seyfi et al. 2015). However, the use of such a method for membrane application has not been reported elsewhere.

3.2.3 Chemical vapor deposition

In CVD, a thin solid film is deposited onto a membrane substrate by means of chemical reactions of the gaseous species (Xu and Yan 2010). This method can deposit coating uniformly over substrate contours and complex surfaces and have higher flexibility of using a wide range of chemical reagents. Superhydrophobic coating with a high WCA ($>160^\circ$) and low SA ($<5^\circ$) was successfully prepared by one-step CVD and modification method using vinyltrimethoxysilane (VTMS), TEOS, and ammonia (Rezaei et al. 2014b). It was shown that the simultaneous CVD and silica nanoparticle modification could coat complex geometries with a superhydrophobic layer. Zheng et al. (2009) modified PVDF membrane by using CVD method from methyltrichlorosilane (MTS) solution. Before CVD process, the PVDF membranes were treated with oxygen plasma to produce active groups that will react with MTS forming a polymethylsiloxane layer. The modified PVDF membrane surface was constructed by microreliefs covered nanometer-scale peaks. The lotus-leaf-like PVDF membrane has a high WCA of 155° and a low SA of 4° , exhibiting superhydrophobic and self-cleaning features.

3.2.4 Sol-gel method

Sol-gel processing is often used to synthesize gels and nanoparticles. This process involves hydrolysis and polycondensation of a precursor, which convert it into a glassy

material (Celia et al. 2013). The surface roughness can be controlled by adjusting the system conditions and reaction mixture (Chao-Hua et al. 2010). Superhydrophobic coatings have been prepared through sol-gel process using metal alkoxide precursors including tetraethyl orthosilicate (TEOS), tetraethyl orthotitanate (TEOT), and tetra-*n*-propyl zirconate (TPOZ), and α,ω -trialkoxysilane-terminated PFPE as a low surface free energy material (Taurino et al. 2008).

Sun (2010) successfully fabricated superhydrophobic PES membrane via sol-gel method using TEOS as the precursor and dodecafluoroheptyl-propyl-trimethoxysilane (DPT-12) for the fluorinated solution. The resultant membrane had WCA of 154° , which was attributed to the micro-texture on the membrane surface combined with the low surface energy. Ahmad et al. (2013e) have fabricated superhydrophobic alumina membranes through dip-coating into sol-gel followed by grafting with (hepta-decafluoro-1,1,2,2-tetra hydrodecyl) triethoxysilane. The time of dip-coating was optimized to promote the appropriate thickness. Further treatments with optimum steam impingement duration and water treatment temperature were performed to create the highest surface roughness and porosity. Superhydrophobic alumina membrane with high WCA of 158.4° and low resistance of $139.5 \pm 24.9 \text{ Gm}^{-1}$ was obtained using dip-coating time of 7 s, steam impingement duration of 1 min, and water treatment temperature of 100°C . The sol-gel method has also been used by Su et al. (2012) to modify inorganic (ceramic) membrane, followed by surface modification with polyurethane-polydimethylsiloxane. The resultant membrane exhibited superhydrophobicity and superoleophilicity with high WCA of 161.2° and low kerosene CA of 0° .

In the sol-gel process, covalent bonds between the coating and substrate are formed by condensation or/and dehydration in the curing process (Chao-Hua et al. 2010). Therefore, a stable superhydrophobic coating can be obtained using this method. Besides, the sol-gel coatings usually have good thermal resistance.

3.2.5 Immersion grafting (direct grafting)

Direct grafting method using hydrophobic molecules is one of the surface modification methods that are commonly used to render hydrophobicity of an inorganic membrane. Grafting occurs by condensation reaction between the OH groups on the inorganic membrane and the functional group of silanes. In typical procedure, the inorganic membrane is immersed in FAS solution for a given period of time under near moisture-free condition with argon atmosphere

to avoid polycondensation process of FAS (Krajewski et al. 2006, Lu et al. 2009, Kujawa et al. 2015).

Cerneaux et al. (2009) prepared superhydrophobic ceramic membrane (Zr50 and Ti5) via grafting with triethoxy-1H,1H,2H,2H perfluorodecylsilane (C8). The grafting efficiency was proven by measuring the LEP_w and WCA, in which the LEP_w and WCA of the C8-g-Zr50 membrane were 9 bar and 160° , respectively. A similar result was obtained by Khemakhem et al. (2013), who prepared superhydrophobic membrane via grafting C8 molecules onto zirconia membrane. After grafting process, WCA of this membrane was increased from 20 to 160° . Lu et al. (2009) fabricated FAS-grafted superhydrophobic alumina membrane, and the alumina membrane was eroded by NaOH solution prior to grafting process. By controlling the eroding time, concentration of FAS solution, grafting time, and grafting multiplicity, superhydrophobic membrane with high WCA of 164.53° was obtained, which is highly attributed to the formation of needle-like surface.

3.2.6 Two-step surface modification

In the two-step method, the superhydrophobic polymeric membrane is prepared by roughening a surface followed by hydrophobization using low-surface-energy material and vice versa. Razmjou et al. (2012) prepared superhydrophobic membrane by increasing the surface roughness through nanoparticles TiO_2 deposition on microporous PVDF membranes using low temperature hydrothermal (LTH) process and subsequently reducing the surface energy through surface fluorosilanization using 1H,1H,2H,2H-perfluorododecyltrichlorosilane (FTCS). The results showed that after modification, the WCA and LEP increased from 125° and 120 kPa to 166° and 190 kPa, respectively. The other approach was dip-coating or immersion of the membrane in a polymer solution containing SiO_2 nanoparticles followed by modification with fluoroalkylsilane (Park et al. 2016, Xu et al. 2017, Zhang et al. 2017).

Liao et al. (2013b) have fabricated superhydrophobic PVDF nanofiber membranes by electrospinning followed by a surface modification on entire nanofibers or only on the membrane surface. After activation with dopamine, the membrane surfaces were deposited with silver nanoparticle and the hydrophobized with 1-dodecanethiol (C12). Compared with pristine PVDF nanofiber membrane, the integrally modified PVDF (I-PVDF) and surface-modified PVDF (S-PVDF) membranes have higher WCA of $\sim 15^\circ$ and $\sim 20^\circ$, respectively. Fang et al. (2011) have prepared superhydrophobic electrospun polyacrylonitrile (PAN) nanofiber membranes by surface coating of silica

nanoparticles and FAS. It was found that increasing particle loading led to higher surface roughness and WCA. When the silica nanoparticle loading was higher than about 15% of the weight of the membrane, the resultant coating surface after FAS treatment became superhydrophobic. The WCA could reach as high as 156.3° and SA decreased to 19.8° . It was shown that at a particle loading of 14.3%, the surface roughness was 310 nm, which was 110 nm higher than that of pure PAN nanofiber membrane. Preparation of superhydrophobic nanofiber membrane through electrospinning followed by controlling the surface roughness and treatment with low-surface-energy fluorosilane was also reported in Hu et al.'s (2016) work.

Madaeni et al. (2013) have prepared superhydrophobic nanofiltration membrane by multiwalled carbon nanotubes (MWCNTs) deposition on the PVDF microfiltration membrane surface followed by PDMS coating. MWCNTs with concentration of 0.05 wt% were used to induce an appropriate roughness. PDMS solutions with different concentrations of 2, 5, and 7 wt% were then used as the hydrophobic coating solution. The WCA of membranes with 2, 5, and 7 wt% PDMS increased to $121^\circ \pm 6^\circ$, $126^\circ \pm 7^\circ$, and $122^\circ \pm 5^\circ$, respectively. This result shows that higher concentration of PDMS leads to improvement of membrane hydrophobicity, but further increase of PDMS concentration can increase the thickness of coating layer, reducing the effects of MWCNTs on roughness and hydrophobicity.

4 Separation properties of superhydrophobic membrane

4.1 Antiwetting for enhanced mass transfer in gas-related membrane application

Superhydrophobic membrane highly repels water or aqueous solution, allowing its pores unoccupied by liquid. Based on Laplace-Young equation (Kim and Harriott 1987), high CAs result in high LEP. This is due to the driving force (transmembrane pressure difference) needed against the capillary pressure [see Himma et al. (2016) for more detail]. Thus, pores of the superhydrophobic membrane are not wetted by the aqueous solution until the high LEP is exceeded. As the penetration of the aqueous solution is prevented, the gas-liquid interface on the mouth of the superhydrophobic membrane pores can be maintained (Figure 6A). As a result, the membrane pores are then exclusively occupied by gas phase and high

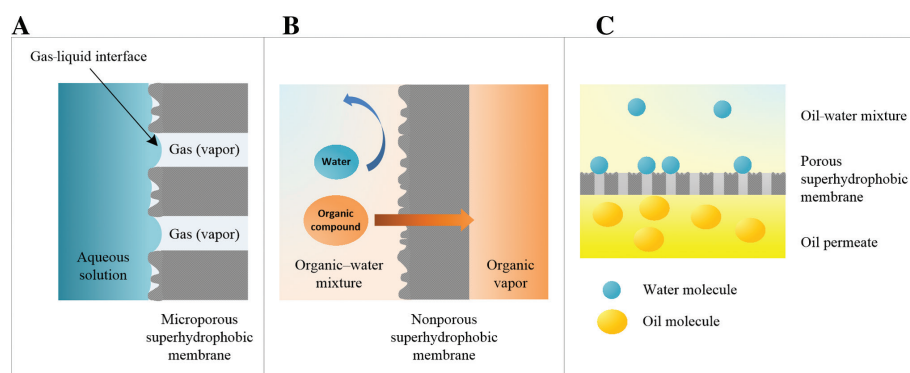


Figure 6: (A) Superhydrophobic microporous membrane exhibiting antiwetting against aqueous solution. The superhydrophobic properties prevent the intrusion of an aqueous solution as the gas-liquid interface on the mouth of the membrane pores is maintained, thus only gas or vapor phase is allowed to transport through the membrane. (B) Superhydrophobic dense membrane exhibiting organic permselective and (C) superhydrophobic and superoleophilic membrane exhibiting superwetting towards oil. The superhydrophobic properties means the membrane has higher affinity to weak-polar molecules than to strong-polar molecules. Thus, the organic molecules or oil are selectively permeated through the membrane.

gas mass transfer rate can be obtained. This antiwetting properties against aqueous solution are highly desired for MGA and MD.

As shown by some studies, the lower mass transfer rate is observed in the less hydrophobic membrane (Nishikawa et al. 1995, deMontigny et al. 2006, Khaisri et al. 2009). This is due to the mass transfer resistance given by the liquid in the pores of membrane leading to loss in long-term performance (Wang et al. 2005, 2013, Zhang et al. 2008b, Franco et al. 2009). According to simulation results by Wang et al. (2005), if the membrane pores were 5% wetted, the overall mass transfer coefficient might be reduced by 20%. Even when below 2% of the membrane pores were liquid-filled, the membrane resistance could be as high as 60% of the total mass transfer resistance (Rangwala 1996). The partial wetting could result in a significant drop of CO_2 transfer rate to 59% of initial value after 14 days of operation time (Lv et al. 2012b).

The use of superhydrophobic membrane can improve the performance of MGA, as summarized in Table 2. From the data, it can be pointed out that the use of the modified membrane with superhydrophobic surface resulted in higher gas flux compared with the unmodified membrane. This could be attributed to the decrease in wetting degree with the increase of hydrophobicity. Lin et al. (2009) showed that hydrophobicity enhancement by CF_4 plasma treatment decreased the wetting ratio of PP membrane by about 74%, 42%, and 55% when using 2-amino-2-methyl-1-propanol (AMP), piperazine (PZ), and methyldiethanolamine (MDEA), respectively, as sole absorbent. In addition to high flux, the gas flux stability also becomes the main concern. This corresponds to the stability of superhydrophobic properties imparted, hence showing

the effectiveness of the superhydrophobic modification. A promising result is shown by Zhang et al. (2012). They showed that hydrophobic modification of PAI membrane through grafting with octadecylamine resulted in a stable flux of CO_2 over 15 days of operation time. Besides, Lin et al. (2009) reported that CF_4 plasma-modified PP membrane experienced a low drop in flux of 4% after 30 days of operation time with a relatively high CO_2 flux.

However, a study by Lv et al. (2012a) showed that superhydrophobic PP membrane prepared by rough polymer film coating exhibited a reduced CO_2 mass transfer rate within the first 7 days, due to the additional mass transfer resistance imparted by the coating layer and the decreased surface porosity. Thus, improvement of antiwetting properties while maintaining the gas mass transfer rate is highly desired.

In MD, the antiwetting properties of superhydrophobic membrane only allow vapor molecules to permeate through the membrane pores, leading to a high vapor flux. Besides, to achieve high flux, wetting should be avoided because the liquid phase occupying the membrane pores will provide direct bridges between feed and permeate (Dumée et al. 2013), resulting in a significant loss of selectivity, which is indicated by increased permeate electrical conductivity (Gryta 2005). It has been noted that in order to prevent wetting, the maximum pore size in the MD membrane should be 0.1–0.6 μm (Alkhudhiri et al. 2012) and smaller pore size ($<0.45 \mu\text{m}$) is recommended for vacuum MD (Tijing et al. 2014), allowing a sufficiently high LEP. However, it should be noted that a trade-off does exist, as reducing pore size may also result in low permeability. MD using a dense membrane could avoid membrane wetting but result in low permeate flux

Table 2: Superhydrophobicity of modified membranes and their separation properties in CO₂ absorption.

Membrane/modification method	WCA (°)	LEP _w (kPa)	Feed gas	Absorbent	Operation time	CO ₂ flux (10 ⁻⁴ mol/m ² ·s)	Increase of flux ^d (%)	Drop in flux ^e (%)	Reference
PP HFM/roughened PP film coating	159	–	CO ₂ /N ₂ (20/80)	MEA (1 M)	20 days	1.9	–18	34	lv et al. (2012a)
PVDF HFM/roughened LDPE film coating	152	–	CO ₂ /N ₂ (20/80)	MEA (1 M)	240 h	31.2	9	57	Ahmad et al. (2013b)
PP HFM/plasma treatment with CF ₄	143	–	CO ₂ /N ₂ (15/75)	AMP (1 M)	30 days	74	8	4	Lin et al. (2009)
PP HFM/plasma treatment with PTFE	–	–	CO ₂ /air (14/86)	MEA (20 wt%)	165 h ^b	9.0	–10	8	Franco et al. (2011b)
					46 h ^c	35.9	27	9	
PVDF HFM/blending with SMM	96.2	800	Pure CO ₂	Water	135 h	3.4	27	12	Mansourizadeh et al. (2014)
PVDF HFM/blending with SMM	92	300	Pure CO ₂	Water	150 h	4.6	142	7	Rahbari-Sisakht et al. (2012b)
PVDF-HFP HFM/grafting with FS10/TEOS	127.8	221	Pure CO ₂	Water	–	6.1	25	–	Wongchitphimon et al. (2011)
PAI HFM/grafting with octadecylamine	105.3	379	Pure CO ₂	Sodium taurinate (2 M)	15 days	17.8	–	~0 (stable)	Zhang et al. (2012)
PEI-fSiO ₂ -NPs HFM/incorporation of fSiO ₂ -NPs	Adv. 123.3 Rec. 107.2	–	Pure CO ₂	Sodium taurinate (2 M)	60 days	150	–	14	Zhang and Wang (2014)
Ceramic membrane (alumina tube with ZrO ₂ layer)/grafting with FAS	153	550 ^a	CO ₂ /N ₂ (12.5/87.5)	MEA (5 wt%)	7 days	12.5	–	26	Yu et al. (2015)
Ceramic HFM (kaolin-alumina)/grafting with FAS	142	248	Pure CO ₂	Water	–	1800	–	–	Abdulhameed et al. (2017)

HFM, Hollow fiber membrane. ^aThe liquid entry pressure of 5 wt% MEA solution. ^bAbsorption of CO₂ with gas stream in the lumen. ^cAbsorption of CO₂ with gas stream in the shell. Other than is gas stream in the lumen. ^dCompared with unmodified membrane; the value is estimated from the available data. ^eCompared with initial flux; the value is estimated from the available data.

Table 3: Superhydrophobicity of modified membranes and their separation properties in desalination using membrane distillation.

Membrane	d_p (μm)	ε (%)	δ (μm)	WCA ($^\circ$)	LEPw (kPa)	MD configuration	T_f/T_p ($^\circ\text{C}$)	C_f (g/l)	J_p ($\text{kg}/\text{m}^2\text{h}$)	R (%)	Reference
PS nanofiber	0.33–1.35	69	60	150.2	60	DCMD	70/20	20	105	–	Li et al. (2014b)
CNT/PVDF-co-HFP nanofiber	0.29	>84	81	158.5	99	DCMD	60/20	35	29.5	99.99	Tijing et al. (2016)
FS-SiO ₂ -PVDF/PVDF nanofiber	0.32	78	72	154	180	DCMD	60/20	35	25	>99.99	Liao et al. (2014a)
GE membrane supported with PET scrim	0.45	80	150	165	–	DCMD	70/20	50	53	>99	Adnan et al. (2012)
PTFE membrane supported with PP non-woven (MS-3020)	0.22	82	160	153	–	DCMD	70/20	50	68	>99	Adnan et al. (2012)
PTFE membrane supported with PET scrim (MS-4010)	0.45	82	140	157	–	DCMD	70/20	50	58	>99	Adnan et al. (2012)
Unsupported PTFE membrane (MS-2000)	0.22	82	30	161	–	DCMD	70/20	50	91	>99	Adnan et al. (2012)
SiO ₂ -PVDF nanofiber (S-PVDF)	0.69	82	102	156.3	175	DCMD	60/20	35	18.1	–	Liao et al. (2014b)
SiO ₂ -PVDF nanofiber (L-PVDF)	0.77	83	129	153.9	179	DCMD	60/20	35	18.9	–	Liao et al. (2014b)
FTCS-TiO ₂ -PVDF membrane	0.42	–	–	163	190	DCMD	70/25	35	30	–	Razmjou et al. (2012)
PVDF nanofiber integrally modified with silver NPs and C12	0.34	77	47	153	146	DCMD	60/20	35	31.6 ^d	–	Liao et al. (2013b)
PVDF nanofiber surface-modified with silver NPs and C12	0.18	68	52	158	86	DCMD	60/20	35	5.4 ^d	–	Liao et al. (2013b)
SiO ₂ /PDMS-coated PVDF membrane	0.36	69	72	156	275	DCMD	90/20	35	5	>99.99	Zhang et al. (2013b)
PVAc grafted CNT/PVDF membrane	–	–	–	157.4	–	DCMD	80/20	35	25	99.5	Zhang et al. (2014)
CF ₄ plasma-modified PVDF membrane	0.25	– ^a	– ^a	162.4	280	DCMD	69/21	40	41	>99.8	Yang et al. (2014)
PVDF-PTFE nanofiber	0.49	69	98	152.2	137	VMD	60/–	35	18.5	>99.9	Dong et al. (2014)
FAS-g-ZrO ₂ membrane	0.2	–	– ^b	142–148	–	AGMD	99/5	58.5	5.4 ^d	~100	Krajewski et al. (2006)
C8-g-Zr50 membrane	0.05	–	–	160	900	DCMD	95/35	29.3	4 ^d	99.8	Cerneaux et al. (2009)
						AGMD	95/20	29.3	4.7 ^d	99.8	Cerneaux et al. (2009)
						VMD	40/20	29.3	7.5 ^d	96.1	Cerneaux et al. (2009)
CF ₄ plasma-modified PSF/PVP membrane	0.6	75–78	–	144.6	260	DCMD	70/20	40	53.33	>99.99	Tian et al. (2015)
C12-g-Ti membrane	0.170	–	–	145	1000	DCMD	90/5	29.3	2.07	99.3	Kujawa et al. (2014)
						AGMD	90/5	29.3	3.86	99.8	Kujawa et al. (2014)
F-POD nanofiber	1.7	–	–	153	90	DCMD	80/22	– ^c	78 ^d	>99.95	Maab et al. (2012)
F-PT nanofiber	2.7	–	–	162	76	DCMD	80/22	– ^c	85 ^d	>99.95	Maab et al. (2012)
PTFE membrane	0.22	53	59	155	759	SGMD	80/21	50	12 ^c	–	Wang et al. (2014)
PDMS/PMMA membrane	8.03	–	65.4	163	13	DCMD	70/25	35	39.61	99.96	Ren et al. (2017)
Methanol phase induced PVDF membrane	0.72	75		165		DCMD	80/20	35	90	>99.99	Munirasu et al. (2017)

d_p , Pore size; ε , porosity; δ , membrane thickness; T_p , feed temperature; T_f , permeate temperature; C_p , concentration of NaCl in feed solution; J_p , permeate flux; R , salt rejection. PVAc, poly(vinyl acetate); C₈, C₈F₁₇(CH₂)₂Si(OC₂H₅)₃ perfluoroalkylsilane molecules; C₆, C₆F₁₃(CH₂)₂Si(OC₂H₅)₃ perfluoroalkylsilane molecules; C₁₂, C₁₂F₂₅(CH₂)₂Si(OC₂H₅)₃ perfluoroalkylsilane molecules; POD, polyoxadiazoles; PT, polytriazoles; AGMD, direct contact MD; SGMD, sweeping gas MD; VMD, vacuum MD.

^aThe porosity and thickness of untreated PVDF membrane were 75% and 125 μm , respectively. ^bThe average thicknesses of ZrO₂ and supporting alumina layer were about 2–3 μm and 25–30 μm , respectively. ^cThe feed solution was real seawater. ^dConverted unit from $\text{l} \cdot \text{m}^{-2} \cdot \text{h}^{-1}$ (assuming permeate density = 1 g/ml = 1 kg/l).

(Purwasasmita et al. 2015). Therefore, improving wetting resistance (increasing LEP) by enhancing hydrophobicity can be an attractive solution. Many studies of MD using superhydrophobic membrane are directed to desalination producing fresh water, which are summarized in Table 3. The data show that superhydrophobic membranes exhibit high rejection of salts (>99%), indicating high efficiency of the modified membrane.

4.2 Enhanced organic permeation and selectivity

Organics have different polarities. It has been recognized that weak-polar molecules have higher affinity to hydrophobic surface than that of strong-polar ones (Dong and Lin 2013). Thus, polar organic liquids such as ethanol and *n*-butanol exhibited lower affinity to hydrophobic membrane than oil, which is non-polar. However, compared with water, which is also polar, ethanol and *n*-butanol have higher affinity to hydrophobic membrane. On hydrophobic PTFE membrane surface, *n*-butanol forms droplets with CA of 26° and penetrates the membrane without any pressure difference, whereas water exhibited high CA of 106° and needed minimum pressure difference of 12 psi to penetrate the membrane (Kim and Harriott 1987). Due to the water repellency and organophilic properties, the hydrophobic membrane has then been widely studied in organic-water pervaporation (Lipnizki et al. 1999, Kujawski 2000, Dong and Lin 2013, Fu et al. 2014, Kujawa et al. 2015, Vandezande 2015). In this case, superhydrophobic membrane eases organic molecules to get

adsorbed onto the membrane surface (Figure 6B) and thus allowing high organic permeability and separation factor (Liu et al. 2011a).

Liu et al. (2013) used hydrophobic nanoparticle incorporation technique to enhance membrane hydrophobicity and organophilicity. They suggest that rendering membrane with superhydrophobicity will reduce the affinity to water but enhance the affinity to organics. In other words, superhydrophobic modification increases organophilicity. Zhou et al. (2014) have shown that the use of the superhydrophobic membrane in ethanol separation from aqueous solution gave separation factor nearly fivefold compared to the membrane without superhydrophobicity. Superhydrophobic self-assembled monolayer (SAM)-modified ZIF-8/PDMS hybrid membrane prepared by Li et al. (2014a,b,c) exhibited a high total flux (1041 g/m² h) and a high separation factor of 95.8 at a feed temperature of 60°C, which are both higher than that exhibited by hydrophobic PDMS membrane. Superhydrophobic MCM-41@ZIF-8/PDMS also exhibited organic (alcohol) permselective due to high water repellency and high alcohol affinity (Wang et al. 2015). Separation properties of the superhydrophobic membrane are shown in Table 4. As the removal of organics from aqueous solution is becoming more important in industrial processes, the development of superhydrophobic membrane exhibiting superior permselectivity to organic compounds will be more attractive.

As oil is a non-polar organic, which has a surface tension of 20–30 mN/m (Zhu and Guo 2016), a hydrophobic material with low surface energy is also easily wettable with oil (liquid with low surface tension). Thus, superhydrophobic membrane designed by combining surface

Table 4: Superhydrophobicity of modified membranes and their separation properties in organic-water pervaporation.

Membrane	WCA (°)	Thickness (μm)	Feed solution	Temperature (°C)	Permeation flux (g/m ² · h)	Separation factor	Reference
ZSM-5/PDMS	138.7	10	5 wt% ethanol-water	40	408	14	Liu et al. (2011b)
SiO ₂ /PDMS	152.0	–	5 wt% ethanol-water	60	244	6.4	Zhou et al. (2014)
MCM-41@ZIF-8/PDMS	144.0	3	5 wt% ethanol-water	70	2204	10.4	Wang et al. (2015)
MIL-53/PDMS	138.3	3	5 wt% ethanol-water	70	5467	11.1	Zhang et al. (2015)
PDMS ^{CF3}	110.0	1	5 wt% ethanol-water	60	412.9	13.1	Tanasescu et al. (2013)
			5 wt% <i>n</i> -butanol-water	60	1292.8	27.3	
ZIF-8/PDMS ^{CF3}	152.4	–	3 wt% <i>n</i> -butanol-water	60	1339	84.8	Li et al. (2014a)
			1 wt% <i>n</i> -butanol-water	60	1041	95.2	
ZIF-8/PDMS	136.8	0.8	1 wt% <i>n</i> -butanol-water	80	4846.2	81.6	Fan et al. (2014)
ZIF-7/PDMS	–	20	1 wt% <i>n</i> -butanol-water	60	1689	66	Wang et al. (2016)
PDMS	122.3	30	1.5 wt% <i>n</i> -butanol-water	55	670.2	43.7	Li et al. (2013)
FAS- <i>g</i> -Al ₂ O ₃	162.0	–	0.6 wt% EA-water	40	282	93.9	Song et al. (2003)
			0.6 wt% PA-water	40	526	129.4	
			0.6 wt% BA-water	40	661	209.9	

BA, Butyl acetate; EA, ethyl acetate; PA, propyl acetate.

chemistry and surface roughness can also have superoleophilic properties (superwetting towards oil). F-SNF/ Al_2O_3 membrane prepared by Huang et al. (2013) exhibited superhydrophobicity with high WCA of 161° and superoleophilicity with oil contact angle (OCA) of 0° . Zhang et al. (2013a) have also successfully prepared superhydrophobic and superoleophilic PVDF membrane with WCA and OCA of 158° and $<1^\circ$, respectively. Membranes with such wettability can be effectively used in oil-water separation, i.e. separation of W/O emulsion (Zhu et al. 2014, Zhu and Guo 2016), in which water droplets are dispersed in the continuous oil phase. Superhydrophobicity allows oil phase to penetrate into the membrane while it repels water (Figure 6C), leading to high separation efficiency. On the other hand, as oil-water emulsions separation is based on size-sieving mechanism, certain pore sizes should also be necessarily maintained to get an optimal membrane design (Si and Guo 2015). Studies of superhydrophobic and superoleophilic membranes and their separation properties in water-oil separation are summarized in Table 5. It can be pointed out that membranes with both superhydrophobicity and superoleophilicity exhibited high oil flux of more than 99% separation efficiency (water rejection) in water-in-oil emulsions separation. Even for surfactant-stabilized water-in-oil emulsions, superhydrophobic and superoleophilic exhibited high flux of $700\text{--}1000\text{ l m}^{-2}\text{ h}^{-1}$, which is still higher than that of commercial UF membrane (Zhang et al. 2013a). Despite the capability of superhydrophobic and superoleophilic membrane in oil-in-water mixture separation (Su et al. 2012), oil yield is very low, i.e. 25–68%. Membranes having superwetting towards oil could potentially be applied in vegetable oil or waste frying oil processing (de Moraes Coutinho et al. 2009, Tres et al. 2010, Tur et al. 2012, Purwasasmita et al. 2013), allowing high oil flux and high rejection of polar compounds and fatty acids.

4.3 Antifouling for non-aqueous and gas-related membrane application

Fouling is the main drawback of membrane application (Fane et al. 2009, Guo et al. 2012, Khoiruddin et al. 2014a, Wenten and Khoiruddin 2016b). Several strategies have been proposed to mitigate the fouling phenomenon, such as improving pretreatment step, operating membrane in appropriate condition (e.g. flux, cross-flow velocity), cleaning, and developing antifouling membrane (Sheikholeslami 1999, Wenten and Widiassa 2002, Rana and Matsuura 2010, Jiang et al. 2017, Wardani et al. 2017). It has been recognized that the surface properties of a

membrane highly affect the membrane fouling tendency. Some approaches have been proposed to investigate the effect of membrane surface wettability on the interaction between foulant and membrane (Azeredo et al. 1999, Choo and Lee 2000, Zhao et al. 2013). Membranes with greater hydrophilicity are less prone to fouling by the organic solute (Goosen et al. 2005, Khoiruddin et al. 2014b, 2017, Aryanti et al. 2015, 2016, Wenten and Khoiruddin 2016a, Ariono et al. 2017). Thus, it can be said that enhancing the membrane surface hydrophilicity should improve the antifouling property towards organic solute in aqueous filtration.

Meanwhile, membranes with highly hydrophobic surface should exhibit antifouling property towards hydrophilic substances in nonaqueous filtration. In addition, the superhydrophobic surface has also been recognized for its self-cleaning ability (Bhushan et al. 2009, Xue et al. 2010). The low surface energy of superhydrophobic membrane should provide the least resistance to removal of fouling (Brady and Singer 2000). This would be favorable for membrane cleaning and flux recovery in any membrane processes.

In real practical application, superhydrophobic modifications have a great prospect to be applied for organic solvent nanofiltration (OSN) membrane. It has been reported that a superhydrophobic membrane resulting from chemical modification using a fluoroalkylsilane compound exhibited more effective fouling resistance than the hydrophobic one in ethanol solvent nanofiltration with total flux decline ratio (DRt) of 23.8% and 29.8% for HA and bovine serum albumin (BSA) foulant, respectively (Guo et al. 2016).

This antifouling property is also useful for gas-related membrane application such as MGA and MD. Different from conventional pressure-driven membrane filtration, membrane contactor such as MGA and MD may not suffer from severe fouling because of the absence of convective flow through pores of the membrane. However, in industrial applications of MGA, gas and liquid streams containing suspended particles can cause pore plugging. Therefore, some studies used superhydrophobic membrane to mitigate membrane fouling in membrane contactor. Yu et al. (2015) reported that superhydrophobic ceramic membrane showed a better anti-fouling ability than did hydrophobic PP membrane. The carbon powder on the superhydrophobic membrane could easily be cleaned. In addition, there was no obvious dust on the superhydrophobic membrane after 1-month operation using flue gas from thermal power plant.

In MD development, membrane fouling becomes one of the major concerns, which is highly associated with

Table 5: Superhydrophobicity and superoleophilicity of modified membranes and their separation properties in oil-water separation.

Membrane	WCA (°)	OCA (°)	Feed mixtures	Performances	Reference
FAS ^a -g-alumina	158.4	—	Kerosene-water	Max. flux = 0.117 l/min Water rejection = 99.99%	Ahmad et al. (2013d)
PS-g-CNTs	152	—	Oil (toluene, hexadecane, or dichloromethane)-water	Flux = 5000 l m ⁻² h ⁻¹ bar ⁻¹ Efficiency (oil purity in filtrate) >99.94%	Gu et al. (2014)
PVDF ^b	158	<1	Oil (petroleum, toluene, isooctane or dichloromethane)-water	Flux (surfactant-free) = 1700–3415 l m ⁻² h ⁻¹ Flux (surfactant-stabilized) = 700–1000 l m ⁻² h ⁻¹ Efficiency >99.95%	Zhang et al. (2013a)
F-SNF/Al ₂ O ₃ SS-CNT	161 Adv: 150 Rec: 143	0 ~0	Petroleum ether-water Oil (isooctane or gasoline)-water	Flux (surfactant-stabilized) = 892 ± 50 l m ⁻² h ⁻¹ Efficiency >80%	Huang et al. (2013) Lee et al. (2011b)
VAMWNT	163	0	Diesel-water	Flux (surfactant-stabilized) = 72.4 kg/m ² h Pa Almost pure diesel could be obtained after the 3rd filtering	Lee and Baik (2010)
PAR-b-PDMS	163	0	Dichloromethane-water	Efficiency = 99%	Li et al. (2016b)
F-PBZ/SiO ₂ NP modified PMIA	161	0	Dichloromethane-water	Flux = 3311 l m ⁻² h ⁻¹	Tang et al. (2013)
F-PBZ/SiO ₂ NPs modified CA	161	3	Dichloromethane-water	Oil quickly permeated, meanwhile water was retained above the membranes	Shang et al. (2012)
PU/PDMS-coated ceramic	161.2	0	Kerosene-water	The highest oil yield = 67.7%, with separation time = 370 min	Su et al. (2012)

CA, Cellulose acetate; F-PBZ, fluorinated polybenzoxazine; PAR-b-PDMS, hydroxyl-terminated fluorinated polyarylester polydimethylsiloxane block copolymer; PMIA, poly(m-phenyleneisophthalamide); PS, polystyrene; PU, polyurethane; SS, stainless steel; VAMWNT, vertically aligned multi-walled carbon nano-tubes.

^a(Heptadecafluoro-1,1,2,2-tetra hydrodecyl) triethoxysilane. ^bThe membrane was prepared via modified phase-inversion process using ammonia additive.

Table 6: Summary of studies on membrane fouling mitigation by superhydrophobic modification.

Base membrane	Modification method	WCA (°)	Application	Foulant (feed stream)	Anti-fouling performance of modified membrane	Reference
Ceramic	Grafting with triethoxy-1H,1H,2H,2H-tridecafluoro-n-octylsilane solution	153	MGA	Dust (Flue gas from coal-fired power plant)	After 1-month operation, no obvious dust was found on the membrane surface, exhibiting a better anti-fouling ability than the PP membrane	Yu et al. (2015)
PVDF	Coating with TiO ₂ and FTCS	166	DCMD	HA and CaCl ₂ solution	The FRR was 94% whereas it was 59% for virgin membranes	Razmjou et al. (2012)
PVDF	Spraying a mixture of PDMS and hydrophobic SiO ₂ nanoparticles	156	DCMD	NaCl solution	After operation, the nanoscale roughness was still maintained, and almost no big particle was observed on the membrane surface	Zhang et al. (2013b)
PVDF	Coating with TiO ₂ and fluoro-silane compounds (FTCS and PDTS)	161 ^a 164 ^b	DCMD	NaCl, HA, and CaCl ₂ solution	There is less residual HA, and the HA was easily removed, resulting in recovery of the CA up to 99%	Meng et al. (2014)
PVDF	Blending silane-modified TiO ₂ nanoparticles into PVDF casting solution	152	DCMD	Gallic acid solution	The flux decline was low with a relative flux (J/J ₀) around 9.3. The membrane could be easily cleaned under UV light and the FRR >90%	Hamzah and Leo (2016)
CNT	Immersion-induced phase inversion/ sintering method and modified with 1H,1H,2H,2H-perfluorooctyltriethoxysilane	168	DCMD	NaCl, CaCl ₂ , Na ₂ SO ₄ , and NOM solution	The membrane exhibited minimal fouling with only few surface area covered by NOM. Meanwhile, both PP and PVDF membranes were completely covered by fouling layers	Fan et al. (2016)
PVDF-HFP	Electrospinning followed by electrospaying of PDMS	155.4	DCMD	Dyes	After 24 h of operation, with water flushing, the WCA was 99% recovered and the membrane surface was much cleaner than commercial PVDF membrane	An et al. (2017)
PVDF	iCVD of PFDA	157	Static MD	NaCl, CaSO ₄ , silica, and alginate	The membrane was able to maintain the air layer, reducing biofouling (alginate and calcium ions)	Warsinger et al. (2016)
PVDF	Deposition of MWCNTs followed by PDMS coating	126	NF	Whey solution	The FRR for membrane coated with 5 wt% PDMS was 82%, exhibiting superior antibiofouling property	Madaeni et al. (2013)
(PEI/PAA-CSH) _n	Chemical modification using PFTS	156	Nonaqueous organic solvent NF	BSA and HA in ethanol solution	The membrane exhibited more effective fouling resistance than the hydrophobic one. DRT of HA and BSA was 23.8% and 29.8%, respectively. And FRR was 82.4% and 75.8% for the first cycle, and still at 89.6%, 82.9% after third cycle, respectively	Guo et al. (2016)
PS	Sonication dip-coating into hybrid hydrophobic silica nanoparticles/PDMS solution	162	Aqueous NF	HA in water	The membrane showed a significant permeance decrease and bad resistance to fouling in the aqueous system. But the membrane should have good fouling resistance to hydrophilic substances and therefore high removal capacity	Shan et al. (2016)

^aFor PVDF-FTCS; ^bfor PVDF-PDTS.DRT, Total flux decline ratio; EEM, the excitation emission matrix; FAS, fluoroalkylsilane; FRR, flux recovery ratio; HA, humic acid; NOM, natural organic matter; NF, nanofiltration; PDTS, 1H,1H,2H,2H-per-fluorooctyltriethoxysilane; PEI/PAA-CSH)_n, poly(ethylenimine)/poly(acrylic acid) (PEI/PAA) calcium silicate hydrate (CSH) multilayered membrane; PFDA, perfluorooctyl acrylate; PFTS, perfluorooctyltrimethoxysilane.

membrane wetting and mainly determines MD performances. As indicated by some studies, foulant deposition on the membrane surface leads to feed solution-filling in the adjacent pores, thereby membrane wetting occurs (Gryta 2005, 2008). This will be accelerated if the foulant is formed inside the membrane pores (Gryta 2005). The fouling layer on the membrane surface increases thermal resistance (Gryta 2008), while the pore blocking creates thermal resistance as well as hydraulic resistance (Tijing et al. 2015). These result in flux decline. In order to mitigate membrane fouling, a number of researches have been conducted by using superhydrophobic membranes, as summarized in Table 6. Different foulants may result in different effects of fouling on superhydrophobic membranes. In general, superhydrophobic membranes exhibited a better antifouling performances towards salt, calcium, HA, and alginate foulants in MD application with various extents compared to the hydrophobic membrane (Razmjou et al. 2012, Zhang et al. 2013b, 2017, Meng et al. 2014, Fan et al. 2016, Hamzah and Leo 2016, Warsinger et al. 2016, Wang et al. 2017). However, in the case of silica foulant, superhydrophobic membranes performed worse than hydrophobic membrane, which might be due to the silica acid polymerization by dehydration and weak repulsion to the negatively charged colloid (Warsinger et al. 2016). Further studies on the role of superhydrophobicity in MD application seems still to be widely opened to obtain high-performance MD membranes that are suitable for real applications.

5 Challenges of superhydrophobic membrane

Superhydrophobic membranes have been successfully prepared from various materials with different techniques. However, improvements are still required to optimize the overall process.

Improvement of conventional membrane process by controlling the system and process parameters can produce superhydrophobic membrane in a single step. Blending modification with highly hydrophobic polymer during conventional membrane preparation process is the simplest method to render hydrophobicity in polymer membrane, but the compatibility between the modifier and polymer must be carefully considered. In addition, as hydrophobicity will only increase by incorporation of low-surface-energy material, it seems quite difficult to obtain superhydrophobic surface with this method. Electrospinning method is able to prepare superhydrophobic nanofiber membrane

directly without further modification. However, low production capacity and limited reproducibility become the limitations for industrial upscaling of the electrospinning process (Ahmed et al. 2015). This method has also a limitation in producing membranes with various shape.

In surface modification by plasma treatment, very thin and strong coating layer can be obtained. However, low-pressure (vacuum) plasma process is generally required. The major disadvantage of the low-pressure plasma process is the requirement of a vacuum system, which increases the cost of operation (Ebnesajjad 2014). Coating with roughened polymer film prepared using solvent-non-solvent method is simple and has low cost. However, as it is a physical modification, there is an issue of long-term performance because of its stability. The formation of covalent bonds between the coating layer and substrate in sol-gel method and grafting method enables to obtain stable superhydrophobic coatings. Modification with depositing coating layer can give additional resistance, which can reduce the benefit of superhydrophobicity that has been rendered. Hence, providing a thin coating layer becomes important. CVD can produce a uniform thin film, but it is a relatively complex method and requires chemical precursors, which usually have safety and health hazards. Fluorinated compounds have been widely used in hydrophobic modification as it has very low surface energy. However, some fluorochemicals and also organic solvent have potential risks to human health and the environment. Minimization of their use will then narrow the available options relating to chemical compounds. Approaches for superhydrophobic modification, which is fluorochemical-free and solvent-free, are then desired.

As reviewed above, superhydrophobic membranes are produced by combining low-surface-energy material and surface roughness. The requisite to use low-surface-energy material raises a limitation in choosing membrane material. For instance, the ceramic materials are very good in mechanical and thermal stability but are not hydrophobic. Another case is that the low-surface-energy material, which acts as the modifier, should be strongly bonded to membrane base material. Thus, a way of superhydrophobization, which is independent of the material hydrophobicity, might be an attractive solution. Liu and Kim (2014) have fabricated superhydrophobic surfaces without using a low-surface-energy materials, which is based solely on surface roughness. Different materials, either hydrophobic or hydrophilic materials, have been tried, and super-repellent surfaces have been successfully obtained.

Meanwhile, for practical application, there are two main challenges that need to be addressed. Producing

a robust superhydrophobic membrane is imperative to maintain satisfactory long-term membrane performance. For MGA application, this means that the modified membrane should be able to withstand high pressure and temperature operating condition. Whereas for MD application, the modified membrane should have high resistance towards fouling and scaling tendency. Apart from this, it is also necessary to minimize pore size reduction that happens during membrane modification. This is crucial to keep membrane flux as high as possible.

6 Conclusion

Rendering membrane with superhydrophobicity is an attractive way to improve membrane performance in many membrane-based separation processes. A surface is called superhydrophobic when its WCA is more than 150° with low CAH and SA of less than 10° . However, the two latter ones are rarely investigated in superhydrophobic membrane preparation.

Numerous methods of superhydrophobic membrane preparation by combining hydrophobic material and surface roughness have been proposed. Superhydrophobic membrane can be prepared by direct processing methods including improvement of phase separation process, blending, and electrospinning. These methods are commonly used for hydrophobic polymer materials. Among these methods, electrospinning is reported to be the method that can produce a membrane with highest WCA. Superhydrophobic membrane can also be prepared by surface modification of the existing membrane, either hydrophobic or hydrophilic membrane. Plasma treatment, roughened polymer film coating, and CVD are a one-step surface modification that simultaneously increases roughness on the membrane surface and incorporates hydrophobic material. Two-step surface modification can also be an alternative method by surface roughening followed by coating with low-energy material or vice versa. Nanoparticles and fluorinated alkyl silane are most widely used for surface roughening and surface hydrophobization, respectively. Up to now, superhydrophobic membranes with WCA over 160° have been successfully prepared.

A microporous superhydrophobic membrane having antiwetting properties against aqueous solution allows the membrane pores to be exclusively occupied with gas/vapor phase, which is favorable for MGA and MD applications. Meanwhile, in pervaporation membrane, superhydrophobicity greatly enhances organic affinity as well as water repellency. This results in high permeability and high separation factor of organics from aqueous solution.

Superhydrophobic membranes also exhibit superoleophilicity, leading to a high efficiency in water-in-oil emulsions separation. In addition to wetting/antiwetting properties, superhydrophobic membranes also have antifouling properties, which is beneficial for OSN application. These promising results should then open new potential applications of superhydrophobic membrane that have not yet been explored including membrane crystallization, membrane emulsification, membrane evaporative cooling, and vegetable oil processing.

However, some challenges are still needed to be addressed for its industrial application. Stability of superhydrophobic surface, membrane thickness optimization, loss in pore size and porosity, and durability in harsh conditions are still to be a concern in superhydrophobic membrane development. Besides, issues of material cost, process simplicity, and environmental aspect should also be taken into account when preparing superhydrophobic membrane.

Nomenclature

γ_{LV}	liquid/vapor surface tension
γ_{SV}	solid/vapor surface tension
γ_{SL}	solid/liquid surface tension
θ	contact angle
r	ratio of the wetted surface area to the projected surface area below the droplet
θ_r	contact angle on a rough surface
θ_e	Young's equilibrium contact angle
θ_c	heterogenous contact angle
f_1	fraction of liquid area in contact with the membrane
f_2	fraction of liquid area in contact with air in the porous membrane
θ_{cr}	critical intrinsic contact angle
θ_A	advancing contact angle
θ_R	receding contact angle
α	sliding angle

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